YoungPyo Hong

Master of Science in Astronomy, Seoul National University
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PERSONAL DATA

- **Birth**: Jun. 4th, 2001, in the Republic of Korea
- **Nationality**: Republic of Korea
- **Family Status**: Single
- **Language**: First language Korean, Excellent in English (TOEFL: 82 / IELTS: 6.0)

EDUCATION

- **Bachelor of Science in Physics** Mar. 2020 - Feb. 2023
  Yonsei University, Mirae Campus *1 year Early Graduated*
  - Admitted as the Top Student
  - Graduated First in the Department
  - ***GPA: 4.05 / 4.3***
- **Master of Science in Astronomy** Sep. 2025 - Aug. 2027
  Seoul National University
  - ***GPA: ? / 4.3***

RESEARCH INTEREST

- Time-Domain Astronomy (Transients – Gamma-Ray Bursts, Kilonovae, Gravitational Wave Events)
- Multi-Wavelength Astronomy & Multi-Messenger Astronomy
- Applications of Machine Learning and Artificial Intelligence in Astronomy

RESEARCH EXPERIENCES

- **Research** Nov. 2021 - Fab. 2022
  A Proposal of Remote Radio Observation in Gwacheon National Science Museum
  - Topic: "*Comparison Between Distance of 'Radio-loud' Quasar and Its Radio Power*"
  - Concluded that No Significant Radio Signals were Detected after Analyzing 18 hours of Observation Data.
- **Post-Baccalaureate Internship** Sep. 2024 -
  Department of Physics & Astronomy, Seoul National University
  - Advisor: **Prof. Myungshin Im**
  - Conducting Research on 7DT Projects Focused on Transient Phenomena and MMA.
- **Papers**

  Gregory S. H. Paek, Myungshin Im, Mankeun Jeong, Seo-Won Chang, Martin Moonkuk Hur, **YoungPyo Hong**, Sophia Kim, Jaewon Lee, Dongjin Lee, Seong-Heon Lee. (2025). "*GECKO Follow-up Observations of the Binary Neutron Star–Black Hole Merger Candidate S230518h*". The Astrophysical Journal, 981(1), 38.
- **Posters**

YoungPyo Hong, Myungshin Im, HongJae Moon, Ji Hoon Kim, Changsu Choi. (2025). "*Filter Transmittance Characterization for 15 New Medium-Band Filters of the 7-Dimensional Telescope: Measurement Procedures and Validation Results*". IASS, Osong, Korea.

INTERNATIONAL CONFERENCES

- J. Cho, T. H. Kim, H. R. Jeon, J. S. Noh, Y. S. Jung, J. H. Choi, C. R. Kim, S. W. Jung, Y. J. Son, **Y. P. Hong**, S. Lim, H. Park, Y. Hahn, E. Ahn, J. E. Kwak, Y. Lee, B. W. Sohn, T. Jung, J. H. Kim. (2022) *"Performance and Application of the Radio Telescope in Gwacheon National Science Museum"*. IAUGA, Busan, Korea.
- **YoungPyo Hong**, Donggeun Tak, Myungshin Im, Ji Hoon Kim, Seo-Won Chang, Gregory S.H. Paek, Hyeonho Choi, Donghwan Hyun, Won-Hyeong Lee. (2026) "*Multi-band Optical SED Analysis of GRB 251013C Afterglow with the 7-Dimensional Telescope*". KAS Spring, Suwon, Korea.
- **YoungPyo Hong**, Myungshin Im, Hyeonho Choi, Sophia Kim, Gregory S.H. Paek. (2026) "*Automated Wide-Field Rapid Follow-up System for Transient Events with TCSpy & gpPy*". IASS, Yeosu, Korea.

AWARDS AND HONORS

- **Yonsei Special Scholarship**, Yonsei University, Korea — Mar. 2020
  - Full Tuition
  - Admission with the Highest Distinction
- **National Program of Excellence in Software Scholarship**, Yonsei University, Korea — Mar. 2021, Sep. 2022
- **Truth Scholarship**, Yonsei University, Korea — Sep. 2022
  - Half Tuition
  - Internal, Academic Excellence
- **Awarded Honors**, Yonsei University, Korea — Feb. 2022
- **Yonsei Vision Scholarship**, Yonsei University, Korea — Sep. 2022
- **Next Generation of Researchers in Basic Sciences Scholarship**, Seoul National University, Korea — Feb. 2026
- **Honorable Mention**, Department of Physics Academic Seminar, Yonsei University, Korea — Nov. 2022
  - "*Comparative Analysis of Internal Combustion Engine Stability Using LEGO Models*"
- **Honorable Mention**, Sci & Tech Convergence Academic Seminar, Yonsei University, Korea — Dec. 2022
  - "*Aerosol Analysis and Precipitation Prediction Based on Changes in Celestial Light*"

EXTRACURRICULAR EXPERIENCES

- **KIAS Outer Galaxy and Cosmology Summer School**
  - 2023: Ran **Ramses & Eunha2** Simulation (Hydrodynamics) / Performed **SED Fitting** using **EAZY**
  - 2024: Ran **GOTPM** Simulation (N-body) / Executed Data Processing for **CMB** data using **CAMB** and **HEALpy**
- **Summer / Winter School on Numerical Relativity and Gravitational Wave**
  - 2022 (Summer): Learned the Fundamentals of **Gravitational Wave** – Principles, Detection, and Data Processing
  - 2023 (Winter): Analyzed Gravitational Wave Data to Determine Q-transform, SNR, and Key Parameter using **PyCBC**, **GWpy**, and **Bilby**
  - 2024 (Winter): Processed Gravitational Wave Data to Investigate Matched Waveforms, Mass, Distance, and Signal Characteristics using **PyCBC**, **GWpy**, and **Bilby**.
- **Radio Summer School**
  - 2022: Studied Basics of **Radio Astronomy** – Receiver System, Interferometer, Galaxy, AGN, and SKA

- **K-GMT Summer School**
  - 2023: Developed a **Proposal** for the **Gemini** Telescope titled;
    "Further Insight on Galaxy rotation curves at High redshifT with GNIRS - FIGTH with GNIRS"
- **Astro AI Asian Network (A3N) Summer School**
  - 2024: Studied **Statistical Modeling** and **Shallow ML** – Bayes, LASSO, Gaussian Process, and SVM
    / Basic **Deep Learning** – MLP and MNIST
    / **Symmetries** and **Specialized Architectures** – CNN, Interpreting CNNs, and Graph Neural Networks
    / **Generative Model** – AE, VAE, GAN, Flow-based model, and Diffusion Model
  - 2025: Studied **Statistical Modeling** and **Shallow ML** – Bayes, LASSO, Gaussian Process, and SVM
    / Basic **Deep Learning** – MLP and MNIST
    / **Symmetries** and **Specialized Architectures** – CNN, Interpreting CNNs, and Graph Neural Networks
    / **Generative Model** – AE, VAE, GAN, Flow-based model, and Diffusion Model
- **ALMA proposal preparation workshop**
  - 2025: Developed a **Proposal** for the **ALMA** titled;
    "Environment of merger events with short GRB afterglow at sub-mm."
- **ALMA Summer School**
  - 2026: Learning how to handle ALMA data and get polarization information.
- **Other Workshops**
  - SKA-Korea Workshop: 2024, 2025
  - Korea Rubin Science Workshop: 2025
  - Woojeon Workshop: 2025, 2026
  - LSST with AI Workshop: 2025
  - Workshop for the 10th Anniversary of the Discovery of Gravitational Waves: 2025
  - Extragalactic Transient Workshop: 2026 (Online)

ADDITIONAL INFORMATION

- **The Korean University Knowledge Mentoring Program** Apr. 2021 - Jan. 2022
  - Provided Academic Tutoring, Motivation-building, Career Counseling, and Caregiving to Elementary, Middle, and High School Students during School Vacations and the Academic Term.
- **The 6th Generation of The Youth Politics Academy** 2022
  - Attended a Program Focused on Educating Young People About Political Systems and Leadership. Participated in Lectures, Discussions, and Activities Aimed at Understanding Governance and Developing Leadership Skills.
- **Astronomy Communicator Level-II** 2022
  - Completed a Program by the Korean Amateur Astronomical Society, Developing Skills in Astronomy Communication and Public Outreach. Trained to Present Complex Astronomical Concepts in an Accessible Way to Non-Experts.
- **Physics Tutoring Activities for University Students** Mar. 2022 - Jun. 2022
  - Provided Tutoring to University Students in Freshmen, Enhancing Their Understanding of Complex Topics.
- **Republic of Korea Army ROTC** Feb. 2022 - Sep. 2022
  - Participated in the Republic of Korea Army ROTC program, Focusing on Leadership and Tactical Training.

Gained Experience in Team Coordination, Tactical Planning, and Field Exercises. Developed Decision-making Skills in High-pressure Environments while Learning Military Operations and Logistics.

- **Department of Physics Student Council** Sep. 2022 - Aug. 2022
  - Served as a Member of the Student Council, Contributing to Student Representation and Organizing Academic Events.
- **Served in the Republic of Korea Army** Apr. 2023 - Jul. 2024
  - Completed Mandatory Military Service, Developing Leadership, Teamwork, and Problem-solving Skills. Gained Hands-on Experience in Challenging Environments while Contributing to National Defense Efforts.
- **The 6th Generation of Dosan Academy Aegiaeta Leadership Program** Aug. 2024 - Dec. 2024
  - Engaged in Leadership Development Based on the Philosophy of "Love Others as You Love Yourself," Inspired by Changho Ahn. Focused on Cultivating Leadership Qualities Centered on Honesty, Integration, and Service to the Community.
- **TA Program in Seoul National University** Sep. 2025 - Dec. 2026
  - Man, and the Universe (2nd semester, 2025)
  - Astronomical Observation and Lab. 2 (1st semester, 2026)